European Elites: An Exploration of LLM Methodologies for Surveying Elite Networks

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Abstract

We present a methodology for LLM-assisted research that anchors every claim to a specific sentence in a source document. The paper situates this methodological intervention within the diffuse field of elite research and the contested question of where power lies in the European Union. Using European institutional elites as a case study—commissioners, DGs, and MEP leaders across seven commission terms (1995–2029)—we demonstrate a pipeline that acquires and hashes controlled sources, segments them into sentences, uses LLMs to propose phrase-level claims, locates every quote deterministically, and submits uncertain evidence to a second adjudicator. The current build contains 5,707 claims, of which 4,326 are verified; 229 remain under review and 1,152 were rejected. The verified subset supports descriptive findings: commissioners concentrate in a small set of transnational policy organisations (ECFR, Friends of Europe, Bilderberg, Trilateral Commission), MEPs in parliamentary and party structures, and DGs in the College of Europe; education is organised around parallel institutional tracks rather than a single pipeline. We report on failure modes encountered—LLM hallucination, non-deterministic extraction, entity duplication, and the limits of string-similarity matching—and describe the methodological choices that address each.

1 Introduction

1.1 Elites in social theory

One of the signature features of the research agenda driven by structural-demographic theory has been its emphasis on elites, and how they figure in the dynamics leading to societal destabilisation, crisis and sometimes collapse. Needless to say, SDT researchers have not been the first to single out elites for social scientific attention. Indeed, there is little doubt that in one way or another, elites have represented an area of considerable interest for students of society and politics since the very dawn of this domain of human inquiry, as attested by Plato's argument for government by philosopher-kings and Aristotle's detailed commentaries on the relative merits and demerits of oligarchy, aristocracy and democracy.

However, in the era of modern social science, elites have been the subject of more dedicated and sustained attention since at least the time of the Italian elite theorists of the late nineteenth and early twentieth centuries—authors from whom we have inherited concepts such as the "organised minority" (Mosca), the "circulation of elites" (Pareto), and the "iron law of oligarchy" (Michels) [2].

Subsequent generations of social scientists have looked at elites from somewhat different angles again. From the mid-twentieth century onwards, studies such as those of C. Wright

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Mills and G. William Domhoff asked searching questions about who truly governed, and about where the agency behind crucial decisions truly lay in polities where elected officials co-existed with powerful unelected (and often non-public-facing) elites. These arrangements were conceptualised in different ways: Mills emphasised the overlap between political, military and corporate elements in the "power elite" that shaped US governance [6]; Domhoff promoted a class-domination theory of power; Michael J. Glennon described the bifurcation between the outward-facing "Madisonian" institutions and the more obscure "Trumanite" network in terms of "double government"; while Peter Dale Scott spoke of "deep politics" or simply the "deep state". However, what these perspectives had in common was the idea that the reality of power looked very different from the pluralist models associated with leading political scientists such as Robert Dahl.

Meanwhile on a more sociological level, scholars such as Pierre Bourdieu and his many acolytes have focused on the processes (often mediated through elaborate cultural and symbolic codes) through which elites ensure social reproduction across generations.

1.2 Elites and the European project

More recently, political scientists working on both sides of the Atlantic have charted various routes through which political elites have seemed to increasingly insulate themselves from the cut and thrust of what used to be considered the basic operations of everyday democratic politics, leaving "cartel parties" and a hollowed out political "void" in their wake (Mair). Some of this research has been focused on the national level, while some of it has honed in on the role of supranational institutions as key sites in the distancing of political power from democratic checks and balances as traditionally understood. Some of it has charted the emergence of a new supranational European elite on the back of the institution-building processes, as Verzichelli and colleagues document [1]. Perhaps the best-documented example of this purported phenomenon is that of the European Union and its component organs, which has spawned an extensive scholarly literature and become something of a byword for democratic deficit—at least in the eyes of its critics, and sometimes, somewhat grudgingly, even in the eyes of its supporters. Much of this EU-oriented literature has been focused on the workings of the European institutions—sometimes casting them explicitly as venues where elite-level decision-making can be hived off from the democratic mechanisms notionally built into national domestic politics.

This has dovetailed with recent research by a number of political scientists based in the US who have deployed fine-grained quantitative research tools to interrogate that polity's purported democratic credentials, only to find that US government policy appears to be shaped overwhelmingly by elite preferences, while the preferences of non-elites seem to exert negligible influence (Gilens and Page). Though not explicitly framed by the researchers in terms of "elite theory", their work is clearly pertinent in this regard.

Comparative work on European power elites likewise finds pronounced social immobility [2].

Taken together, all of this might seem to indicate the existence of a thriving ecosystem of elite research, whatever its implications for the socio-political health of contemporary societies. However, the relative abundance of research pertaining in one way or another to elites belies a remarkably diffuse field of enquiry. Across this landscape of elite-focused studies, we find research questions that are not just distinctly different from one another, but sometimes in certain important respects pulling in opposite directions. Some of these questions concern the perpetuation of elite dominance. Others, arguably, are about something close to its inverse. Even within the narrower field of specifically SDT-driven research, we find distinctly different kinds of questions about elites (and would-be elites) being formulated. Some of these are concerned with what elites try to do (often unsuccessfully) to stave off crisis and collapse in tumultuous times. Others are concerned with how the overproduction of elites or elite aspirants directly *contributes* to societal destabilisation. This latter seam of research can be split, in turn, into at least two rather distinct substrata. To put it somewhat crudely, this means looking not

just at elites as "winners", but at elites (or perhaps more accurately "would-be", "wannabe" or "has-been" elites) as "losers". Both phenomena can be highly destabilising. As Turchin has pointed out, the overproduction of the elites that we might call "winners" tends to result in the ratcheting up of the "wealth pump", leading to unsustainable levels of extraction from the wider economy and society, and the gradual impoverishment of non-elites. Meanwhile, the overproduction of "elites as losers" poses its own problems, which can take a number of forms. One example might be that of the *ancien régime* French nobles who lashed out against the crown's attempted reforms in the run-up to the crisis of 1789, seeing them as only being likely to speed up the trajectory of downward social mobility they perceived themselves to be on, thereby unwittingly undermining the remaining supports of the system that they clung to, limpet-like, for sustenance, and ultimately accelerating France's slide into revolution and civil war. Another example might be the heavily indebted graduates of US law schools and other advanced degree programmes who are currently trying and failing to turn their credentials into lucrative careers, and increasingly breaking away from the traditional political formations to support more radical offerings—identified by Turchin and other SDT researchers as a recipe for serious instability.

Thus "elite research" is not one thing, but many things, and any examination of this shorthand immediately throws up fundamental questions, including but not limited to the following: Whom are we really talking about when we talk about "elites"? How many different types of "elite" are there, and who should really be called "elite"? In so far as "elite" is a meaningful concept, how does one attain this status, or maintain it in an era in which many individuals with "elite" status are experiencing downward social mobility and becoming *déclassés*? Relatedly, how does one lose elite status? Do traditional means of attaining or maintaining elite status, such as acquisition of higher education (cf the famous "college premium"), still apply, or is this now being washed away in a sea of credentialism, degree inflation and unsustainable student debt? What is the relation between the "winner" and "loser" elites? Is there a firm distinction between the two, or more of a revolving door?

And perhaps most importantly: how do these various questions relate to the moment of profound and multifaceted crisis that Western societies are currently experiencing?

1.3 Why the European Commission?

To be sure, this represents a tangled skein of questions rather than a strict research agenda, and while we cannot address all of them in our paper, we do propose to at least bring together a number of distinct strands of elite research as they pertain to Europe's current conjuncture. In so doing, we hope to achieve a number of goals. As well as bridging the gap between these somewhat disparate strands of elite theory, we aim to apply the resultant framework to relatively novel subject matter. Previously, SDT-inspired elite research has been applied to various historical and contemporary societies, but never, to our knowledge, to a supranational entity or supranational elite. Something similar could be said of "power elite" research along the lines of Mills or Domhoff. No consensus may exist as to how to explain what happens in the inner sanctums of American power, but at least we have some reasonably firm hypotheses to choose from, and to build upon. How power works at the level of Europe, meanwhile, is notoriously something of a black box.

This is not to draw an equivalence between American and European power. While the EU has long been recognised as a behemoth in areas such as trade, scholars have continuously debated the extent to which it can be considered a coherent political actor in a larger sense. Indeed, some scholars argue that, insofar as the EU is a power on the international stage, it is a declining one. Nonetheless, declining or otherwise, one cannot attempt to make sense of the current conjuncture without attempting to make sense of the EU and the elites that animate it. Whatever one thinks about the efficacy of the EU, it is no longer possible to think about German or French policy, either on a domestic or foreign level, in isolation from European

policy. Nor is it possible to think through the reactivation of Europe as a major theatre in the current not-so-cold war between East and West without considering the EU's role.

Of course, part of the reason that the inner workings of the EU are so obscure is that the institutional architecture of the European project is unusually complex. "Studying the EU" could mean many different things, and the first questions one faces when attempting to do so are "how?" and "where to begin?". Among the various institutions that comprise the EU, we have chosen to focus our initial analysis on the European Commission, as we believe that if any one entity embodies the supranational elite that is emerging on the back of the European project, it is this more or less permanent central bureaucracy. (Not for nothing is the Commission sometimes described as the beating heart of the EU.) However, we acknowledge that this basic starting premise of our analysis is open to a certain degree of debate, insofar as no consensus exists in the scholarly literature as to where power lies within the European project, or even as to the European project's basic nature. These questions have been the subject of many decades of intense debate, and while we cannot make them the focus of this paper, we do need to briefly address their basic outlines, insofar as they pertain to the current analysis.

Put simply, the various schools of thought on the nature of the European project, while manifold, can roughly be divided into two broad camps—that which understands the EU as a fundamentally inter-governmental initiative, and that which views it as an experiment in supranational governance. Furthermore, this difference in perspective is not merely scholarly or theoretical: the EU clearly incorporates both inter-governmental and supranational elements and, what is more, is riven by tensions over this very issue. That is, the theoretical difference in opinion maps onto a real-life one, in that the protagonists of the European project have themselves chronically disagreed over whether it ought primarily to be an intergovernmental or supranational enterprise. And while the actually existing structures incorporate both logics, it is difficult to say conclusively which is more important, as those structures have evolved on the back of multiple compromises, via multiple rounds of treaty negotiations, between actors with rivalling visions as to which logic ought to prevail. Thus this tension is woven into the basic fabric of the European institutions, and snapshots taken at different historical junctures since the founding of the European institutions would show the battle lines drawn in different ways between the intergovernmentalists and the supranationalists, with the initiative periodically slipping back and forth between the two dispositions.

In short, the question of where power lies in Europe is shrouded in some opacity, and we do not need to reach for the scholarly literature to see evidence of this. It is reflected in many of the stock phrases customarily used in real life to describe the workings of the European project. Indeed, when facing criticisms from "Eurosceptic" political actors who lament the eclipse of national sovereignty, their "pro-European" counterparts cheerfully acknowledge that the EU is built on a principle of "pooled sovereignty". But it is not always clear where exactly this sovereignty "pools together", as it were. Indeed, an oft-repeated line widely attributed to Henry Kissinger reflects this opacity. In reference to the diffuse nature of the European power structure, Kissinger is purported to have quipped: "Who do I call when I want to speak to Europe?"

On the other hand, some commentators would suggest that, if US leaders ever really did struggle to "speak to Europe", as Kissinger is said to have suggested, this issue would seem to have been less salient in recent times. From this point of view, Europe has, in spite of increased trans-Atlantic verbal sparring in the Trump era, largely abandoned the independentist streak of the 1990s and early 2000s and increasingly fallen into line with the United States, at least on key foreign policy issues.

To the extent that this is true—and clearly the point is controversial—it raises a further difficult question for anyone attempting to carry out research on the EU elite, and another dimension to the question of where power lies. That is, we are no longer merely pondering whether power really lies in the Commission or in the Council, or in Brussels, Frankfurt, or the

various national capitals. We also must take seriously the question of whether it even lies in Europe.

In sum, the study of "elites" is the study of many different things: rising elites, declining elites, would-be elites, entrenched elites, "power elites" and "elites" who may not yet be obviously experiencing downward social mobility, but whose real power is open to question. In turning our focus to the cadres of the European Commission, we leave open the question of exactly what kind of elite it is. However, taking it that it is an elite of some kind, and that shedding light on how it is constituted would be useful in the current conjuncture, we take a preliminary step toward this goal in the following sections.

1.4 From elite theory to a citation-anchored method

The step we take is methodological. The evidentiary challenge follows from the conceptual one. Elite formation is distributed across 27 member states with distinct educational systems, career pathways, and political cultures; the networks that connect these elites cross national boundaries and institutional sectors [3]; and the relevant evidence is scattered across biographies, CVs, declarations of interest, and institutional records. Capturing it requires aggregating data across languages, jurisdictions, and source types, while keeping every claim traceable to its source.

The remainder of this paper describes a methodology that treats the LLM as an extraction engine, not an authority. Every claim in our database is linked to a specific sentence in a specific document. The LLM's output is a hypothesis, verified against the source before acceptance. The current build contains 5,707 claims, of which 4,326 are verified, across 3,850 entities and 823 hashed source documents. It supports analysis of seven European Commissions from Santer (1995) through Von der Leyen II (2024–2029).

2 Data and methods

LLM-assisted extraction produces candidate statements, and hallucination is one failure mode. Self-consistency checks [5], self-evaluation under controlled formats [4], and citation-grounded generation [7] motivate the design used here. They do not establish that this workflow extracts relations accurately. The method therefore separates three operations: proposing a subject–relation–object candidate, locating a recoverable source span, and deciding whether that span supports the proposed relation. Source location is a necessary condition, not a guarantee that the relation is entailed, correctly dated, or true.

The pipeline has six stages: acquire and hash sources; segment each source into sentences with exact character offsets; propose candidate claims; locate and grade their evidence; adjudicate uncertain evidence; and resolve entities and render the graph. Each stage writes a durable artifact: a source snapshot, a sentence table, claim and evidence rows, assertion decisions, and a graph export.

2.1 Study design and populations

The study population is a set of unique people, not person–term observations: 150 commissioners across seven terms, 85 Directors-General and Deputy Directors-General, 1,000 corporate elites, and 98 judicial elites. MEP records are retained in the corpus but excluded from the analyses below. Each frame has its own roster and source cutoff; a person is counted once within a frame even when several roles or sources refer to the same individual. The frames are comparison groups for descriptive coverage, not a random sample of a wider European elite population.

2.2 System Architecture

The pipeline has six stages: acquire and hash sources; segment each source into sentences with exact character offsets; propose candidate claims; locate and grade their evidence; adjudicate uncertain evidence; and resolve entities and render the graph. Each stage writes a durable artifact: a source snapshot, a sentence table, claim and evidence rows, assertion decisions, and a graph export.

2.3 Database Schema

The database is best read as an evidence chain rather than a table inventory. A source snapshot is segmented into sentences; a candidate claim records a subject, predicate, object and role; one or more evidence rows point to recoverable sentence spans; and an assertion records the acceptance decision together with the evidence grade and adjudication history. Entities, aliases and organisation profiles are supporting registries, not additional evidence. The relational tables implement this chain; the unit of analysis is the claim and its located evidence.

The implementation stores: Acquisition, model-run and graph-query details are recorded in the repository audit directory.

- `source_document`: one snapshot per source, with URL, local path, SHA-256, revision, retrieval date, and access date.
- `sentence`: exact character offsets; sentence IDs, not list positions, are the only locator used downstream.
- `entity` and `entity_alias`: persons, organisations, institutions, commissions, and countries, with aliases for resolution.
- `claim`: a candidate triple—subject, predicate, object, and role. A claim is not yet a finding.
- `evidence`: the claim, source document, sentence span, exact quote, locator status, and flags for subject, object, and role.
- `assertion`: the decision—verified, needs_review, or rejected—with an evidence grade of A, B, C, or none. Only verified assertions enter the graph.
- `adjudication`, `rejection`, and `review`: recorded model verdicts, rejection reasons, and human decisions.
- `organisation_profile`: sourced or manual-seed organisation attributes, kept separate from graph edges.

A verified assertion must have at least one located evidence row. A claim whose quote cannot be found in the cited document is rejected rather than assigned a guessed index. The invariant checker enforces this, together with foreign keys, source hashes, and the absence of duplicate active entities.

2.4 Phrase-Level Extraction

Documents are segmented once into sentences with exact character offsets. The corpus is not truncated: long biographies are chunked, and every sentence keeps its ID. The LLM receives a numbered chunk and returns an organisation, a role phrase, a relation, the exact evidence sentence, and the sentence number. The quote is located by normalised text matching: the locator searches for a normalised prefix of up to 240 characters, progressively shortening it to

40 characters if necessary. A prefix match is recorded as such; the remainder of the proposed quote is not treated as supported. If no normalised match is found, the claim is rejected rather than assigned a guessed index.

A separate grading step records what the located sentence span contains. Grade A means that the subject, object and a role keyword are present in the located span; the subject may be supplied by the document's subject identity. It does not establish grammatical entailment, negation, date validity or factual truth. Grade B means subject and object are present with the role implicit; grade C means a required element is missing and the case is sent to adjudication. In this paper, *verified* names the workflow's acceptance state, not independent confirmation of the proposition.

2.5 Official Source Acquisition

Wikipedia biographies supplied broad initial coverage, but a substantial minority of people had no local source at all. We therefore acquired official institutional documents directly: 26 member biographies from the Court of Justice of the European Union (CURIA), six Commission Director-General and Deputy Director-General CVs or person pages, and five corporate leadership biographies from Eni, SAP, Safran, and Neoenergia. Each document was fetched from its institutional URL, converted to text, hashed, segmented, and attached to its subject; extraction then ran through the same claim, evidence, and assertion path as every other source. These acquisitions are source-resolution decisions documented in the repository; their coverage effect is not used as a before/after experiment in this paper.

Missing sources are recorded as missing evidence, not as negative findings. The acquisition history is documented in the repository rather than used as a coverage result in this paper.

2.6 Entity Resolution and the Limits of String Similarity

Extraction produces organisation names as they appear in source texts: "Radio Netherlands", "Radio Netherlands Worldwide", "FEPS", "Foundation for European Progressive Studies". String similarity alone is dangerous: "CFR" and "ECFR" score 0.857 on difflib, yet one is the US Council on Foreign Relations and the other the European Council on Foreign Relations. An auto-merge rule set at 0.95 similarity would merge them: one false-merge example, not a general threshold recommendation.

The registry resolves names with a deterministic merge key that strips parenthetical qualifiers, normalises punctuation and accents, and preserves sub-entities such as an "EPP group" as distinct from the party. It carries aliases, stable slugs, external identifiers where available, and a quarantine status for junk surface forms. Residual fuzzy variants are listed for review rather than silently merged. Every final merge is reproducible from the registry.

2.7 Organisation Profiles

Organisation type, funding, headquarters, and descriptions are stored in `organisation_profile`, separate from graph edges. The current seed is manually researched and marked `manual_seed`; it is not presented as a sourced assertion. The graph does not use LLM-generated classifications for colouring or analysis. Entity categories come from the registry, and uncertain classifications remain in the review queue.

3 Method assessment and research outputs

The verified layer renders to a browsing snapshot with 3,485 pages: 460 people, 2,494 organisations, 7 commissions, 519 source documents, plus index, statistics, and citation pages. The snapshot is not a count of the audited corpus (994 source documents); pages and graph exports

describe earlier renderings and should not be read as claim or document totals. Each page shows only verified assertions, with role, date, quote, evidence grade, and a link to the source document. The wiki is a browsing interface; the analysis input is the typed graph export, not the Obsidian view.

The typed graph is exported separately: 3,226 nodes and 3,850 edges, of which 2,837 are network edges, 486 education, 431 occupational, 94 institutional, and 2 other. The typed export carries edge class, predicate, role, grade, quote and source location for every edge; the analysis uses that export rather than the browsing view.

Metric	All verified	Grade A only	Network edges only
Nodes	3,226	2,842	2,520
Edges	3,850	3,211	2,837
Unique person–entity pairs	3,633	3,080	2,702
Largest component	2,828	2,395	2,024
Degree-1 nodes	2,670	2,355	2,116

Table 1: Numerical sensitivity of the graph export to evidence grade and edge class. These are export counts, not estimates of extraction accuracy.

The stricter grade-A view retains most edges (3,211 of 3,850) but fewer nodes (2,842 of 3,226); the network-only view retains 2,837 edges. The central network claims should therefore be read as stable in direction under these filters, not as numerically identical.

The network rendering is not reproduced here: at 3,226 nodes it is unreadable at page scale. The typed export and the numerical sensitivity table are the analysis record; the wiki remains the browsing view.

The review layer is explicit. Assertions that the deterministic checks cannot settle are marked `needs_review`; grade-C claims are rechecked by a second model, and disagreements are not silently promoted. In the audited release, 235 assertions remain under review; they are not counted as verified. The queue is exported as a priority-scored table, but the audited database contains no completed human-review rows: the queue describes work still to be done, not a completed validation exercise. Model agreement is an acceptance rule, not independent evidence of correctness. The run metadata in the audited database identifies the extraction and adjudication model as `deepseek-chat`; the second-judge rechecks use `deepseek-flash` with thinking enabled. Prompts and cache keys are implementation artifacts. A ten-case recheck of grade-A claims retained eight and sent two to review (a governing-board versus academy member confusion and a founder relation not stated in the quote). This is a failure-mode sample, not a precision estimate, and no independent human validation was completed.

4 Descriptive findings

The introduction reports a frozen release of 5,707 claims and 823 source documents. The body analyses a later audited release: 6,716 claims, 994 source documents, 5,331 verified assertions, 235 under review and 1,150 rejected. These counts are not interchangeable; all figures below refer to the audited release unless stated. The following findings describe the verified subset, not the larger candidate pool. The dataset covers 150 commissioners, 85 DGs and DDGs, 1,000 corporate elites, and 98 judicial elites. MEP records are retained in the corpus but excluded from the graph. Coverage is uneven: commissioner education is verified for 119/150, DG/DDG for 78/85, corporate for 64/1,000, and judicial for 98/98. The education table below uses the expanded non-MEP frame; the network findings describe the verified graph snapshot. We therefore report descriptive patterns and avoid claims that depend on the unverified remainder.

4.1 Elite Network Structure

Commissioners with accepted affiliations share a policy-institute cluster: the European Council on Foreign Relations (15 commissioners), Friends of Europe (11), Bilderberg (9), and the Trilateral Commission (8). DGs and DDGs centre on the College of Europe (8). In the observed corporate subset, recorded shared affiliation is lower: the largest shared organisation is the European Round Table of Industrialists (3), and 8 of 1,000 share an elite-network organisation with a commissioner under our broad list.

These are counts of distinct people with accepted affiliation records, not simultaneous membership or contact; affiliations are pooled across careers. The contrast is therefore a difference in recorded overlap, not a measured fragmentation statistic, and corporate coverage is much thinner than commissioner coverage.

4.2 Recorded affiliations with transatlantic organisations

The outcome here is a recorded affiliation with an organisation in a manually defined transatlantic category; it is not a measure of influence. The category is an analyst codebook, not a sourced assertion: organisations are grouped into a hard set (Bilderberg, Trilateral Commission, Atlantic Council, European Leadership Network, Munich Security Conference, German Marshall Fund, GLOBSEC, Slovak Atlantic Commission) and a softer set (European Council on Foreign Relations, Friends of Europe). The latter are excluded from the hard count. On the hard set, 20 commissioners have at least one accepted affiliation, accounting for 32 affiliation links; two corporate elites have three links. No accepted DG/DDG affiliation was identified under these source and coding rules.

These counts describe recorded affiliations in the observed subset. They do not establish attendance, participation, policy alignment or causal influence, and a zero indicates no accepted link under the coding rule, not an absence of affiliation.

4.3 Gating Institutions

Table 2 reports the institutions most frequently represented among educated commissioners, Directors-General and deputy Directors-General, corporate elites, and judicial elites. MEPs are excluded. The table uses the expanded non-MEP education frame (1,000 corporate and 98 judicial people); the network counts above describe the prior verified graph snapshot. We use "gating institution" in a deliberately narrow sense: an institution whose graduates are repeatedly present in elite roles. Presence is measurable; necessity is not. The table therefore identifies potentiators, not proven gates.

Institution	Commissioners	DGs/DDGs	Corporate	Judicial	Total
College of Europe	3	8	0	7	18
Harvard University	6	2	0	6	14
University of Cambridge	2	3	0	5	10
École nationale d'administration	3	0	1	6	10
London School of Economics	4	1	1	2	8
UCLouvain	1	2	1	4	8
University of Oxford	3	1	2	2	8
Vilnius University	3	0	0	4	7
École Polytechnique	0	0	7	0	7
University of Cologne	2	0	1	3	6

Table 2: Education by institution and non-MEP elite track. Counts are educated people, not degree rows. Denominators: commissioners 119, DGs/DDGs 78, corporate 64, judicial 98. Corporate education coverage is 64/1,000; judicial coverage is 98/98.

The aggregate is weakly concentrated: the top five institutions cover 56 of 359 educated people (15.6%), and the mean is 1.89 institutions per educated person. The College of Europe has the largest total (18 people: 3 commissioners, 8 DGs/DDGs, and 7 judicial elites) but spans three of the four tracks. The London School of Economics, Oxford and UCLouvain each span all four tracks, with eight people apiece. Harvard has 14 people across three tracks and no observed corporate link in the 64 corporate people with education records; École Polytechnique is corporate-specific (7 of 64) and absent from the other tracks. ENA is concentrated in the political and judicial tracks.

These are descriptive associations, not causal effects. Denominators differ substantially: commissioner and DG/DDG education is much better covered than corporate education, and the judicial frame has complete education coverage because it was built from mandatory official biographies. The pattern supports parallel institutional pipelines rather than a single European elite school.

4.4 Exploratory indicators of educational output and position availability

Structural-demographic theory treats elite overproduction as a mismatch between qualified elite aspirants and the positions available to them. The core question is not who enters a gating institution, but who, having acquired the qualification, converts it into an elite position and who remains an unplaced aspirant. Entry into the gate is therefore upstream of the mechanism; the relevant transition is qualification to appointment. Graduate counts are not production in this sense unless they identify people who are both qualified and seeking elite positions. Our data permit only a partial test of that transition.

This subsection is exploratory. Output trajectories and nomination totals answer different questions, and neither is a complete test of overproduction. The longer series now assembled do not show a uniform expansion. The College of Europe's matriculation grew from 35 in 1950–51 to 494 in 2024–25, with most of the increase before 2000. Harvard's LL.M. degrees moved from 95 in 1983 to 175 in 2024, with a COVID-year trough of 59. École Polytechnique's French-CPGE entrants were 383 in 2002, 393 in 2008, 383 in 2010, 382 in 2014, and 390 in 2019, before the 2020–25 series; the 2002–19 band is flat at 379–397, and the booklet series rises to 431. Oxford's BCL candidates examined rose from 67 in 2002 to 111 in 2019, while MJur candidates fell from 61 to 38. Cologne's comparable first-examination series begins in 2010 (444 graduates), reaches 358 in 2011 and 420 in 2024, with a low of 320 in 2014 and 370 in 2021; the 2007 old-law backlog and the incomplete 2008–09 reports are excluded. LSE's LLM enrolment was 277 in 2014/15 and 273 in 2020/21. ENA/INSP remains a short series: 80 in 2019, 89 in 2024, and 61 in 2025. These trajectories are not a common output measure, but they are long enough to show that elite production is not rising uniformly across the gating institutions. A chained median index of per-capita output, which annualises each observed interval and distributes it across the intervening years, rises from 100 in 1960 to 297 in 2025, with a peak in the early 2000s and a lower plateau after 2010.

Denominator data. Figure 2 isolates the two elite-position denominators used here. A is the European Commission's establishment-plan posts, all grades (AD, AST and AST/SC), permanent and temporary, across all Commission budgets: 23,356 posts in the 2025 budget and 23,357 in 2026. It is not the staff headcount of 33,071: contract agents, local agents, seconded national experts, agency staff and Commissioners themselves are outside this post count. The 2025 value is observed; the 1990–2020 benchmarks are linear interpolations between official all-budget establishment-plan anchors.[1]

[1]Draft General Budget 2027, Working Document II, `https://commission.europa.eu/document/download/307081a9-ad13-4027-943d-dc4b52a2e493_en`.

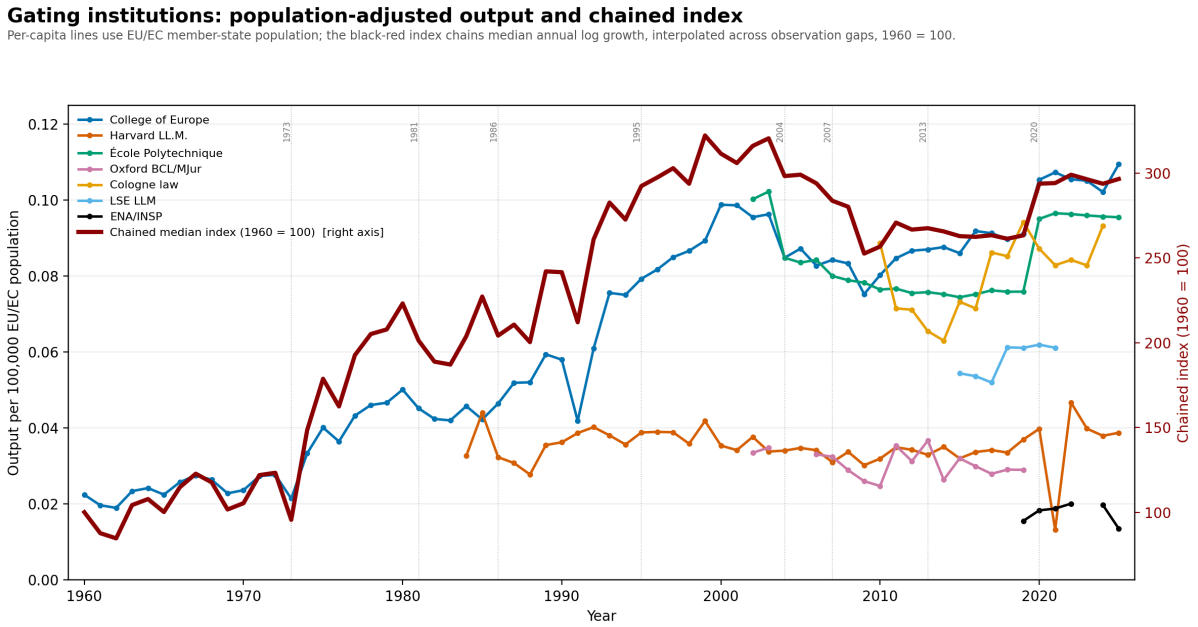


Figure 1: Available gating-institution output per 100,000 EU/EC population, 1960–2025. The coloured series use each institution's own reported output measure; the population denominator is the EU/EC member-state population at each date, with enlargement and Brexit breaks. The dark red line is a chained median index of per-capita growth (right axis, 1960 = 100); each observed interval is annualised and interpolated across missing years, so a new observation does not create a level jump when a series enters or leaves. The population is a contextual scale denominator, not a population at risk of admission or appointment.

B is the number of board seats at publicly traded companies in Germany, France and Italy: 7,771 seats in 2025, down from 16,716 in 2000. It counts seats, not unique people; a director with two mandates contributes two seats. For Germany it counts both the supervisory board (*Aufsichtsrat*) and the management board (*Vorstand*); for France the *conseil d'administration* or *conseil de surveillance*; for Italy the *consiglio di amministrazione*, excluding the separate *collegio sindacale*. Employee representatives are included. The 2010 and 2025 values are the best-observed points; other benchmarks multiply listed-company counts by average board size from national governance studies, with documented interpolation where direct counts are unavailable.[2]

For the Court of Justice of the European Union, Article 255 panel reports distinguish first-term candidates from renewals. Between 2019 and 2022 the panel recorded 50 first-term candidate assessments and 45 openings at the Court of Justice, the Advocates General and the General Court. These are not a matched numerator and denominator: the candidate opinions cover January–February 2022 while the openings cover all of 2022, and opinions have not been linked one-to-one to posts. The 30 renewal opinions in the same period are excluded. We therefore report the two administrative totals separately and do not compute a candidate-to-opening ratio, a conversion probability or a lower bound on competition.

4.5 Recorded post-mandate activities

The extract contains 159 recorded post-mandate decisions: 90 following the Juncker Commission, 32 following the first von der Leyen Commission and 37 from earlier periods. Each is an

[2]FidAR Women-on-Board-Index 2026; HCGE 2025 annual report; Consob 2025 corporate-governance report; construction in `audit/elite_position_denominator_5y.md`.

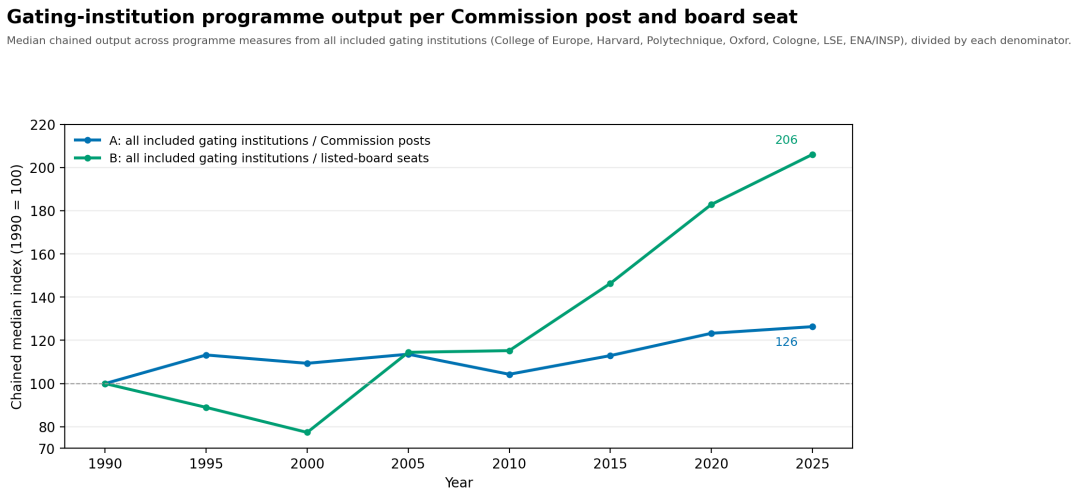


Figure 2: Gating-institution programme output per Commission post and per listed-board seat, 1990–2025 (1990 = 100). Both lines are median chained output indices for programme measures from all included gating institutions—College of Europe, Harvard, École Polytechnique, Oxford, Cologne, LSE and ENA/INSP—divided by the relevant denominator. A and B are stocks of positions/seats, not counts of unique people.

administrative record of an approved activity, not a person-level transition; a person may account for more than one record. By destination, 62 (39%) fall into the residual category "other", 25 are academia, 20 finance or corporate, 14 think tank or advisory, 13 NGO or foundation, 13 consultancy or lobbying, seven international organisations, and five public or political. The size of the residual category limits any claim that destinations are concentrated in the named substantive classes.

A second, narrower measure comes from the DG/DDG tenure layer: 14 post-service intervals and six service intervals. This is a pilot, not a complete senior-official roster. Neither measure can be converted into a revolving-door rate, because we lack a denominator of officials eligible to move and because the Commission records notifications rather than all subsequent activities. The defensible result is the count and classification above, not a frequency or prevalence claim.

5 Discussion and limitations

The contribution is a workflow for recording candidate relations, recoverable evidence and decisions about acceptance. Nothing in the design guarantees that an accepted relation is true: the grade records what appears in the located span, the adjudicator records a decision, and later review can revise either. The useful claim is narrower than reliable extraction: a reader can inspect the source, the quoted span, the decision and the reason for rejection.

Coverage is uneven across the frame. Commissioners and senior administrators have substantially better educational and biographical coverage than the 1,000-person corporate frame; judicial biographies are unusually complete because they come from mandatory official sources. Absent links are therefore not negative observations, and cross-track comparisons describe the observed subsets rather than the underlying populations. The same caution applies to the organisational data: an affiliation is not influence, and co-affiliation across a career is not contemporaneous contact.

The position indicators are exploratory scale comparisons. Commission establishment-plan posts are posts, not incumbent persons; listed-board seats are seats, not unique directors; educational measures include degrees, entrants, enrolment and examined candidates. A flow-to-stock ratio from these components is not an appointment probability. The population series is

a contextual denominator, not a population at risk of admission or appointment; enlargement and Brexit can move it for reasons unrelated to institutional output. With a fixed 2025 EU-27 geography, the 2025 chained index is 540.9 rather than 296.6; the difference is denominator geography, not output. A substantive overproduction test would need matched aspirant and opening definitions, appointment-linked cohorts, mobility and exits.

Future work follows from those limits. The next useful studies are an independently annotated validation of extraction precision and recall; institution-specific reference populations for educational representation; and harmonised applicant, nominee and opening series. Each addresses a different uncertainty. The internal acquisition queue and model operations belong in repository documentation rather than in the research argument.

The reproducible release is the database `euro_sdt_v2.db`, the source snapshots, the analysis scripts and the commit recorded with the manuscript. Re-running extraction or adjudication may require API access and can produce different outputs; reproducing the published tables from the frozen accepted records does not. Model identifiers, prompts, caches and run metadata are kept with the implementation rather than reconstructed from prose.

6 Conclusion

The bibliography was checked with `check_citations.py` as a review aid. The checker extracts citation claims from the LaTeX source, searches a bounded window of the cited text, and asks a model to judge phrase-level support. Its coverage is partial: it evaluates the first 500 segmented phrases and truncates each phrase in the prompt. The committed report has nine rows with confidence 1.00, while the checker prompt uses a 1–5 scale and the repository's stated threshold is 3; that report format predates the current scale and is not evidence that every citation passed the current threshold. Model approval is not independent validation of the cited claim, and absence of support in the searched window is not a full-document result.

The contribution is an inspectable workflow, not a guarantee of accurate extraction. Hashed source snapshots, sentence-level locators, recorded evidence decisions and a review queue make errors visible and revisable; they do not by themselves establish reliability. The case study demonstrates that this discipline yields auditable data and descriptive findings, while also showing which earlier claims do not survive stricter verification. That is the appropriate outcome for a methodology paper: the evidence layer is designed to make such revisions visible.

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